capable of comparing *one-way* light speeds in various directions with sufficient accuracy (1 part in 10^4) to test the Newtonian prediction. However, there was a similar prediction of a difference in *round-trip* light speeds (about 5 parts in 10^9 difference between a round-trip parallel to the Earth's motion through the aether and one perpendicular). Michelson's new technique was ideally suited to measuring such round-trip differences; they were what Michelson searched for and could not find.

65 [By contrast, Heinrich Weber . . . mislead young minds.] I do not know *for certain* that Weber was confident of this, or that he in particular took the attitude that it would be inappropriate to mention the Michelson–Morley experiment in his lectures. This passage is speculation based on the absence of any sign that Weber discussed the experiment, or the issues raised by the experiment, in his lectures; see the detailed notes on his lectures taken by Einstein (Document 37 in ECP-1) and the brief description (page 62 of ECP-1) of the only other existing set of notes from Weber's lectures.

65 [By comparing it with other experiments,] The other experiments were those, such as measurements of the aberration of starlight, which implied that the aether is not dragged along by the Earth; see note to page 63, above.

65 [A tiny (five parts in a billion) . . . Michelson–Morley experiment.] Recall (note to page 64) that Michelson was actually measuring round-trip light speeds and looking for variations with direction of about five parts in a billion.

66 [If one expressed . . . (see Figure 1.1c).] This discussion of the "no ends on magnetic field lines" law, and the more detailed discussion in Figure 1.1, is my own translation, into modern pictorial language, of one aspect of the Maxwell's equations issue with which Lorentz, Larmor, and Poincaré struggled. For a more precise discussion of this issue and their struggle, see pages 123–130 of Pais (1982).

66 [If the Fitzgerald contraction . . . "dilates" time.] To make the laws beautiful required not only the contraction of moving objects and the dilation of their time, it also required pretending that the concept of simultaneity is relative, i.e., that simultaneity depends on one's state of motion; and Lorentz, Larmor, and Poincaré paid considerable attention to this as well as to length contraction and time dilation. However, for pedagogical simplicity, I ignore this in the text and take up the issue of simultaneity somewhat later in Chapter 1.

68 ["I am more and more convinced . . . not correct."] Document 52 in ECP-1; Document 8 in Einstein and Marić (1992).

68 [Over the next six years, . . . dilation of time.] Here I am speculating. It is not really known to what extent Einstein's mind focused on these issues during 1899–1905. As Pais (1982, Section 6b) makes clear, during these six years Einstein was unaware of the Lorentz–Poincaré–Larmor deduction of length contraction and time dilation from Maxwell's laws. Stated more technically, he *was* aware of Lorentz's derivation of the Lorentz transformation up to first order in velocity (including simultaneity breakdown), but not to second order where length contraction and time dilation occur. On the other hand, he presumably was aware of the Fitzgerald–Lorentz inference of length contraction from the Michelson–Morley experiment; and we do know that in his 1905 paper on special relativity he gives his own derivation of the full Lorentz transformation, accurate to all orders, and of length contraction, time dilation, and simultaneity breakdown.

69 [To the saucy . . . Mileva Marić,] For a description of Marić's personality based largely on the love letters between her and Einstein, see Renn and Schulmann (1992); for the love letters see ECP-1 or Einstein and Marić (1992).

69 ["I'm absolutely convinced . . . bad recommendation."] Document 94 of ECP-1;